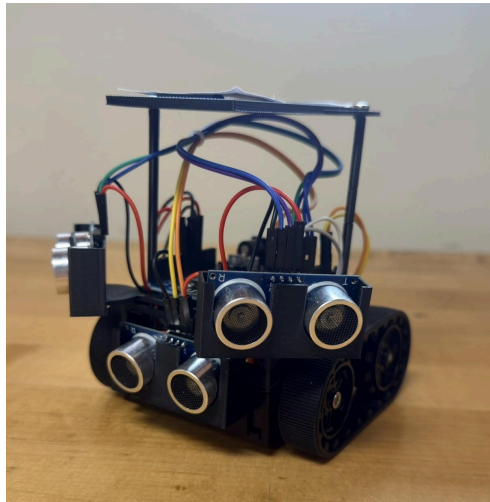


Autonomous Driving Robot Final Report



ENGR122-J1: Field Sustainable Systems with Sensors
Group 3

Jaiden Garcia - Biomedical Engineering
Abilash Ramasamy - Computer Engineering
Aryan Bhatnagar - Mechanical Engineering

“I pledge my honor that I have abided by the Stevens Honor System.” - J.G, A.B,
A.R

Executive Summary - Ramasamy

Project Requirements:

Stevens Institute of Technology has asked our team to design and implement an autonomous mobile robot capable of guiding visitors around campus during tours. Within the context of the arena, the robot is required to autonomously navigate to the four designated target locations using real-time positional data. Specifically, the robot will receive its current X and Y coordinates through the MQTT-based communication network. This will serve as the primary localization input for navigation and control.

The specific functional requirement is that the robot must successfully reach all four target locations on the obstacle course within a strict time constraint of three minutes. To achieve this, the system must implement an efficient path planning and obstacle avoidance strategies which takes into account the static obstacles present within the environment.

Our Final Design Performance:

Our final robot demonstrated efficient and reliable autonomous navigation. It successfully completed the obstacle course and reached all four designated target locations in 51 seconds (our best time). This was well below the three-minute requirement. Over the final two weeks of the project, we iteratively tuned the ultrasonic sensor thresholds and the motor speed parameters. This allowed us to generate a well-calibrated path planning and obstacle avoidance system for our robot. The path planning algorithm produced streamlined trajectories that attempted to minimize unnecessary motion. Additionally, the robot consistently avoided obstacles without any collisions. It maintained appropriate distance from the surrounding objects based on the configured ultrasonic sensor parameters established in our code. Overall, this performance reflects a strong balance between speed and safety. Our robot showcased fast, precise, and dependable autonomous operation within the given constraints.

Introduction - Ramasamy

Overview of Technologies Used:

1. Software:
 - a. SolidWorks: This platform was used to design the robot's 3D body.
 - b. FlashPrint: This platform enabled us to add supports to our designs and prepare them for 3D printing.
 - c. Arduino IDE: This was the platform where we wrote the C++ code for the robot.
 - d. MQTT Network: A communication protocol that is used to receive the real-time position from the overhead camera system.
2. Hardware:
 - a. Pololu Zumo Robot Chassis Kit
 - i. This was the base kit assigned to us. It contained two silicone tracks, two drives, and two idler sprockets.
 - b. Wemos D1R2 (ESP8266) - Wi-Fi enabled development Board
 - i. Microcontroller that handles WiFi connectivity and logic
 - c. 2 Micro Motors with Metal Gear with 2 Pins JST Male Connector
 - i. These were the D3 and D6 servo motors of the robot.
 - d. 3 Ultrasonic Sensor Module (HC-SR04)
 - i. These were used to receive Pulse Width Modulation (PWM) values.
 - ii. They were vital when it came to developing the obstacle avoidance program for the robot.
 - e. Feitech - SMC - 2 Channel (FT-SMC-2CH)
 - i. This was the motor driver which controlled both the servo motors present within the robot.
 - f. OLED module (0.96 inch) 128x64, SS1306 Driver I2C
 - i. This was the screen attached to the robot which displayed relevant values such as the three different values produced by each ultrasonic sensor.\
 - g. 4.8v 3000mah NiMH Battery
 - i. This served as the robot's power source. It had one terminal for the voltage, and another one for ground.

Project Objectives & Design Requirements - Ramasamy

The primary objective of this project was to design and program an autonomous mobile robot which was capable of navigating a structured arena to reach four target locations. The robot needed to use real-time positional data (X and Y coordinates) transmitted through the MQTT network. This would allow it to perform the necessary path planning. The key goal was to make sure that the robot operated autonomously without any external control.

The task of reaching all four designated target locations had to be accomplished within a maximum time limit of three minutes. At each target, the robot needed to stop for at least two seconds indicating that it recognized the target. Additionally, the robot needed to avoid having collisions with any obstacles present within the arena.

The design was constrained by several mechanical, electrical, and software requirements. From a mechanical standpoint, the robot body needed to be created using SolidWorks. Each sketch needed to be 3D-printed using FlashPrint and the Adventure 3 Series 3D-printer. There was a maximum total printing time of five hours for all our 3D-printed components. As far as the purpose, the final design needed to securely mount all three Ultrasonic sensors, the Arduino microcontroller, and the OLED display. Additionally, we could not use any adhesives such as glue or tape. It is important to note that we needed to develop a custom mounting bracket to hold the two ArUco markers. These markers play a significant role in tracking where the robot is in real-time within the arena.

From a software perspective, the robot needed to take into account the camera system along with at least three Ultrasonic sensors to enable accurate localization and real-time obstacle detection. These sensors needed to be positioned strategically in order to increase the robot's environmental awareness. We had to use the PWM values provided by these sensors to develop an obstacle avoidance algorithm. Additionally, we needed to generate an efficient path planning algorithm in which the robot knew where to navigate to. In essence, the final program was a product of both the obstacle avoidance and path planning algorithms.

In terms of electrical design, all components needed to be connected to the Arduino microcontroller. No breadboards were permitted. This meant that we had to use power buses in our design. It was also mentioned that proper color coding was needed for the wires to indicate specific connections. Furthermore, a power switch was also mandatory as a simple way to turn the robot on and off.

Overall, this project required careful consideration of all the objectives and constraints outlined above. Mechanical, electrical, and software decisions had to always align with these requirements in order to produce a viable and complete final design.

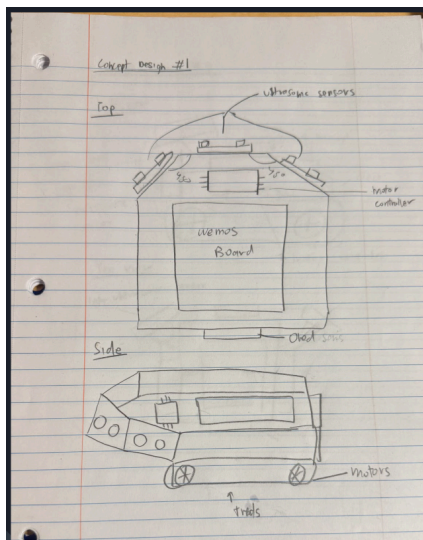
Concept Design & Design Alternatives - Bhatnagar

Design Objectives Morphological Chart:

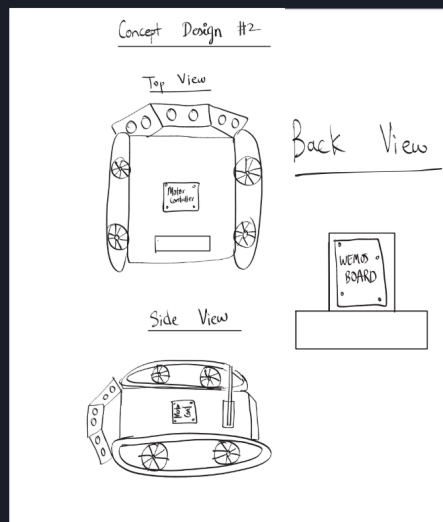
Functions	1	2	3
Sensor Angle Placement	30, 90, 150 (degrees)	45, 90, 135 (degrees)	0, 90, 180 (degrees)
Wemos Board Positioning	Horizontal	Vertical	Slanted
Cable Management	Wire	Plastic twist ties	Zip Ties
Component Management	3D Printed Casing	Plastic Casing	Wood Casing
How to fasten components	Glue	Screws	Zip Ties

Based on the Design Objectives we clarified above, the group decided to create 3 different concept ideas that would implement the objectives above and try to add some different touches to see which one turns out the most unique and meets the criteria.

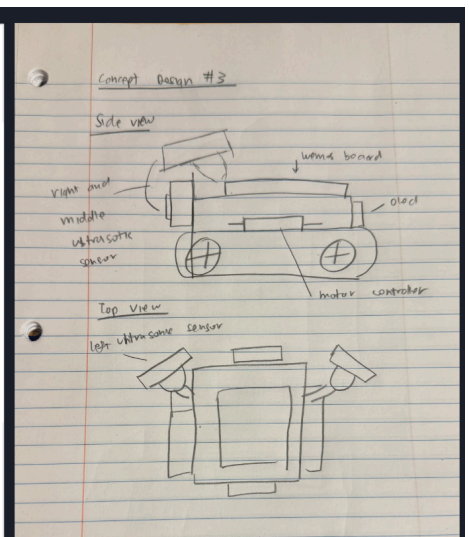
Concept Ideas



Concept #1



Concept #2



Concept #3

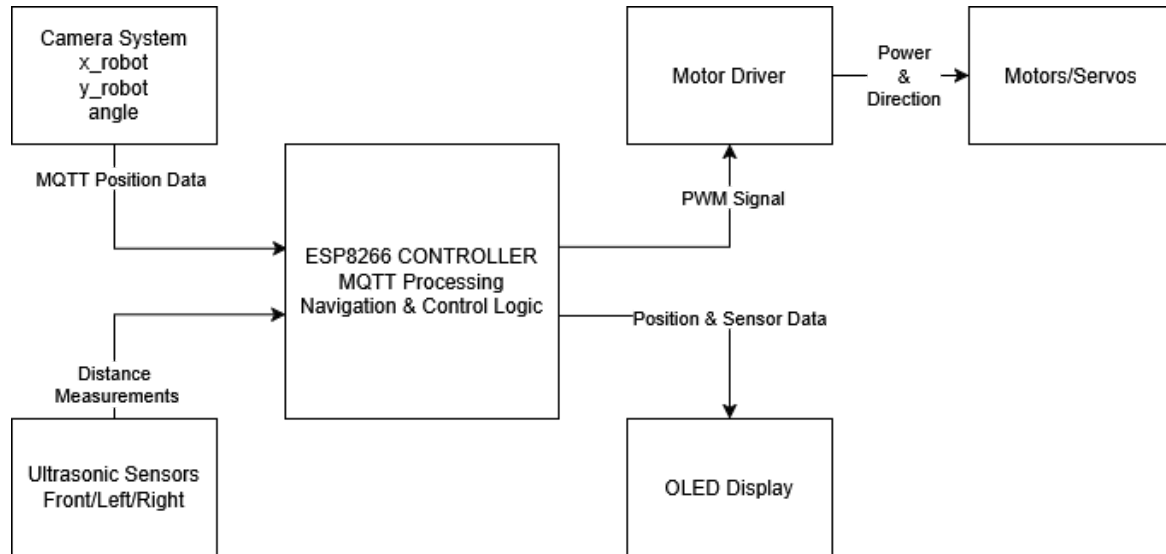
Concept Evaluation Matrix (Lowest Score Wins):

	Concept #1	Concept #2	Concept #3
Size	3	1	2
Durability	3	2	1
Design	3	2	1
Easy to make	1	2	3
Safety	3	2	1
Total Score	13	9	8

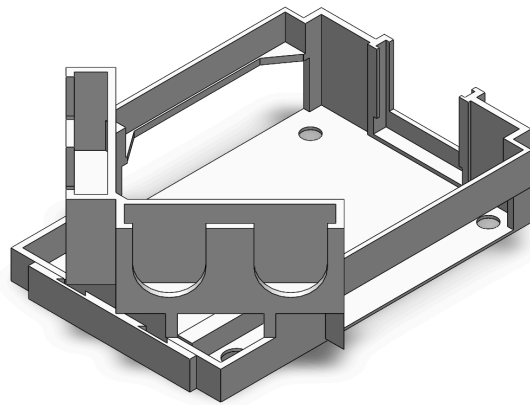
After using the concept matrix, we found out that the most functional and unique concept was concept #3 due to the criteria listed above. The final concept we chose meets the design objectives of the project by giving us an aesthetic and pleasing look at design. The design is nicely mapped out and is very compact. It leaves a lot of room for wires at the bottom, reducing the number of exposed wires. It is fastened by screws providing a lot of durability and structure to the design.

Final Design Description

System Description and Architecture - Garcia

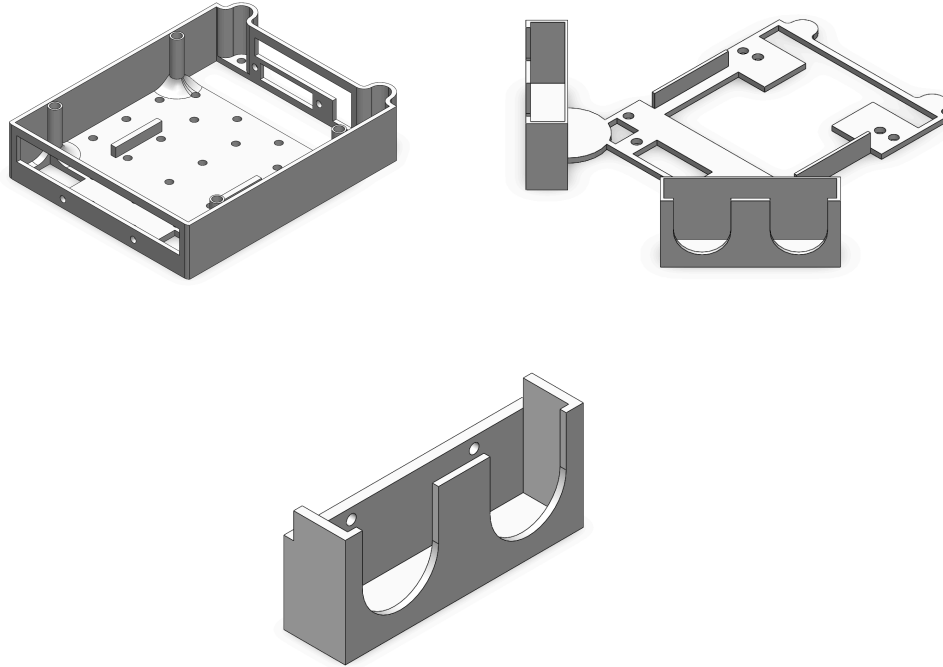


Mechanical Design - Bhatnagar, Ramasamy



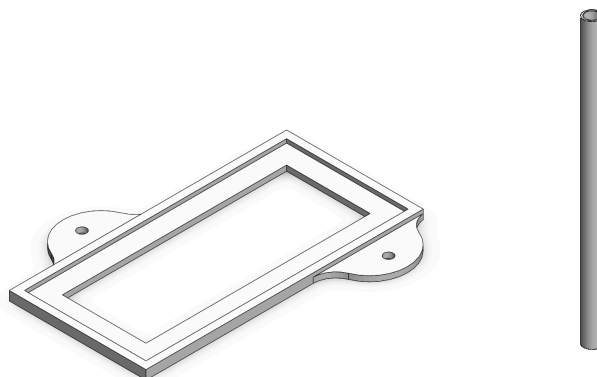
Design #1

We originally created this sketch to place the two Ultrasonic sensors which we needed for the Back and Forth Road Test and the Left-Hand Loop. We also created a slot for the OLED display later on.



Design #2

Design #1 took quite a long time to print. Hence, we scrapped that design and created Design #2 which had three different parts: a sketch for the base, a sketch to hold two Ultrasonic sensors, and a sketch that was a standalone mount for the third Ultrasonic sensor. This design offered us maximum printing time efficiency.

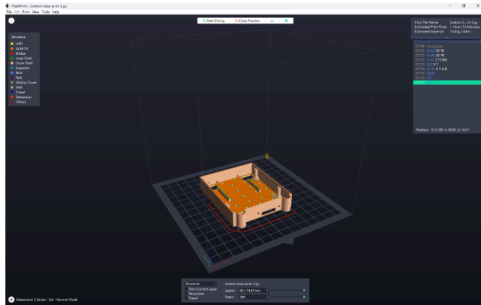


Design #3

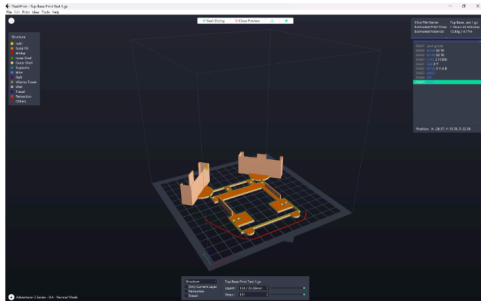
This design was in addition to Design #2. We added two rods and a mounting bracket to accommodate for the two ArUco markers needed on the robot.

Total Printing Time - 4 Hours and 44 Minutes

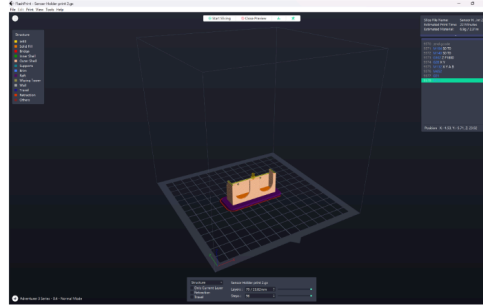
Bottom Base Print Time - 1 Hour 53 Minutes



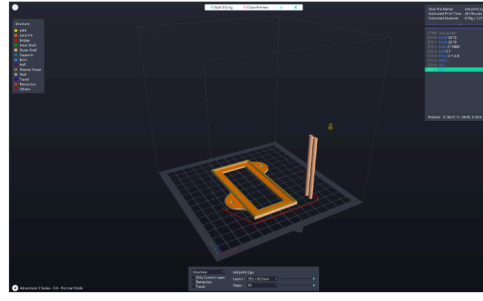
Top Base Print Time - 1 Hour 42 Minutes



Middle Sensor Holder - 22 Minutes

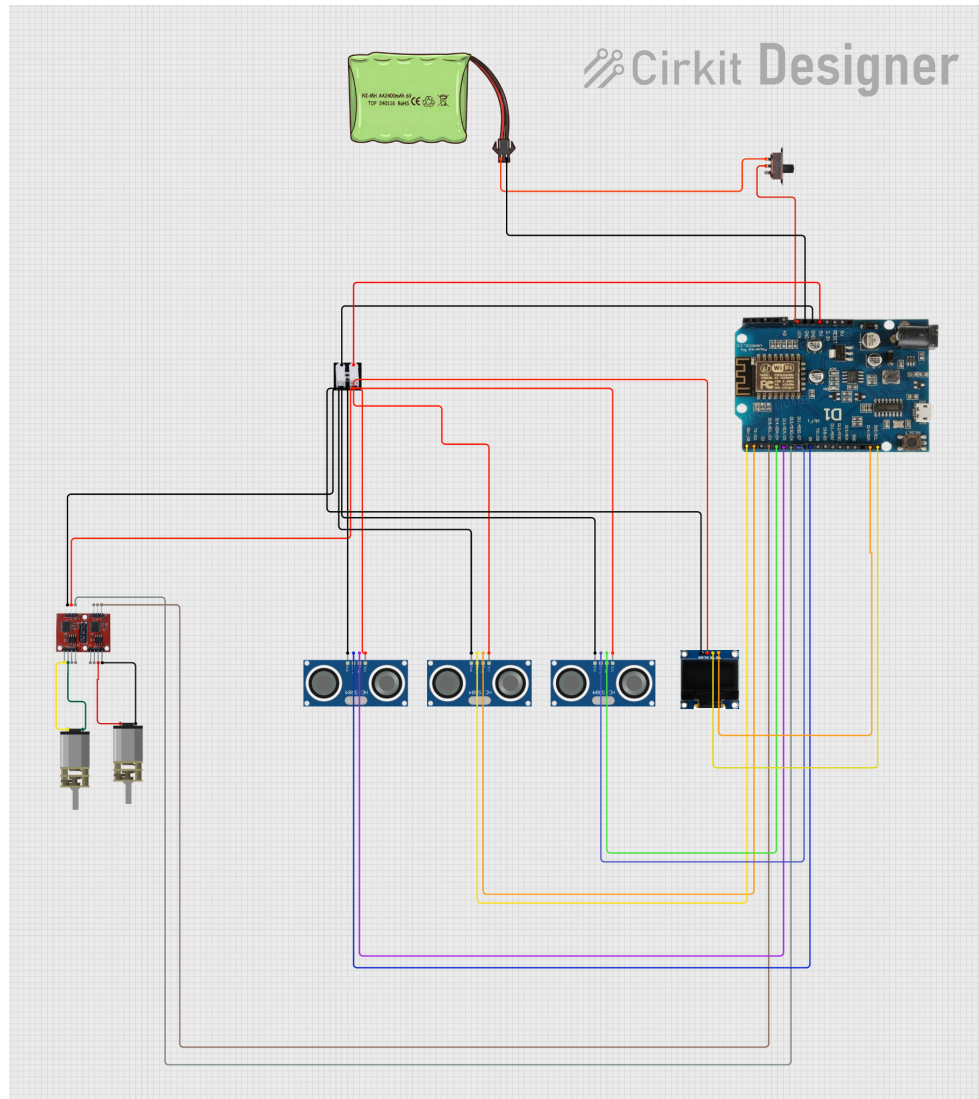


Marker Holder - 47 Minutes



Our total printing time on the Adventure 3M Series was 4 hours and 44 minutes, just short of 5 hours. This design combined a lot of technical solidworks skills and assembly in order for it to get the printing time so low. If we stuck with our original design it would have taken a lot of longer due to the supports.

Electrical & Wiring Design - Ramasamy



Circuit Design made with CirkidDesigner

Color Coding:

- Red —> Voltage
- Black —> Ground

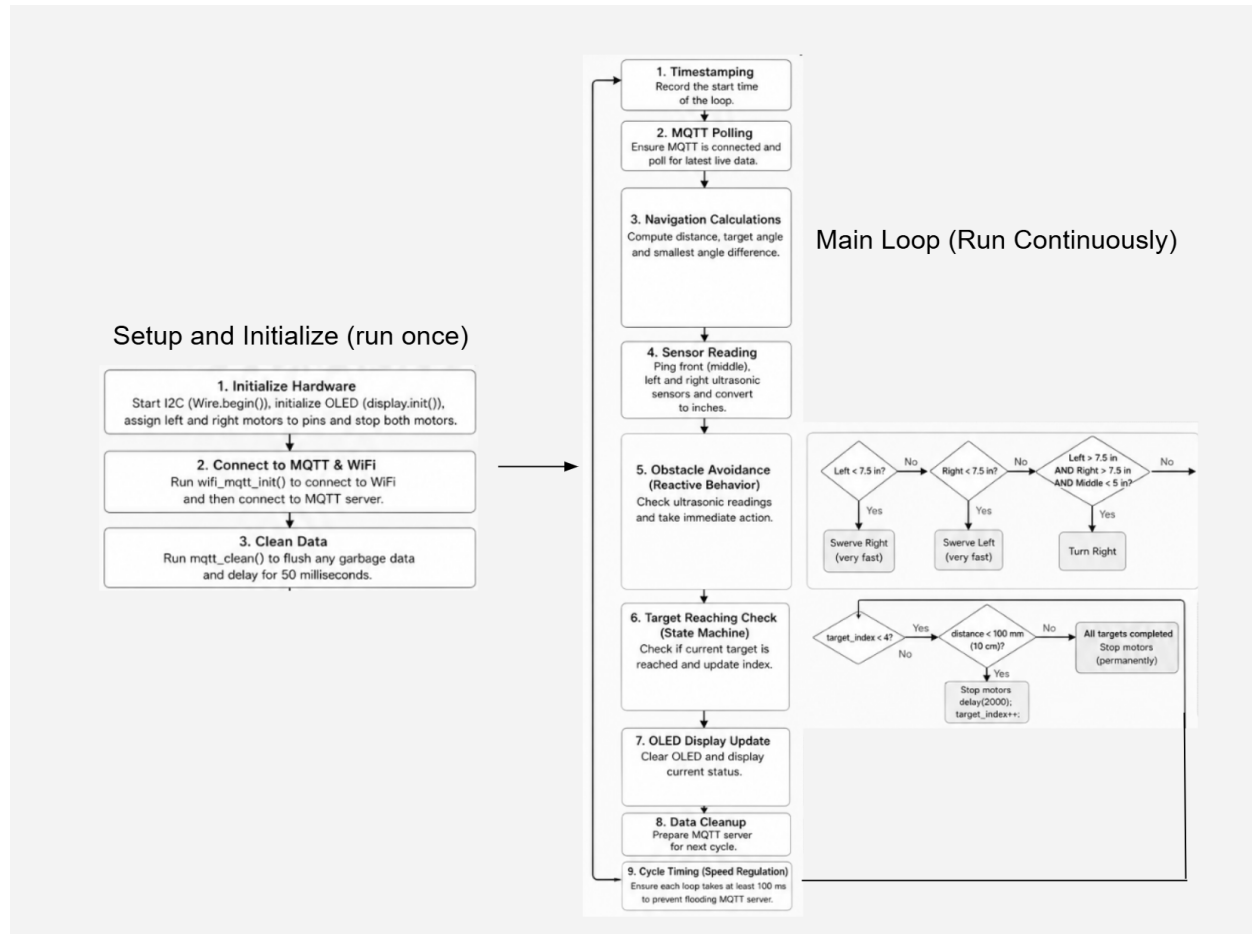
Our electrical design used two power buses for organized power distribution. One bus was designated for voltage. Each ultrasonic sensor, the motor controller, and the OLED display were connected using red jumper wires. Each component's voltage line was routed to the power bus, which was then connected to the Arduino's 5V pin. Similarly, a separate ground (GND) bus was used. All

components shared a common ground connection to the power bus that was ultimately routed to one of the Arduino's GND pins.

For power control, a switch was integrated between the battery and the Arduino board. The positive terminal of the battery was connected to one pin of the switch, while the second pin of the switch was connected to the VIN pin on the Arduino board. The battery's ground terminal was directly connected to the Arduino's GND pin, completing the circuit.

In terms of communication and types of signals used, multiple protocols were used to interface with the system components. The ultrasonic sensors depended heavily on the digital GPIO pins. These were the pins where trigger signals were sent and echo signals were received to calculate distance. The motor controller functioned by using PWM (Pulse Width Modulation) signals from the three Ultrasonic sensors on board the robot. This allowed for precise control of motor speeds and the robot's direction. Additionally, the OLED display communicated with the Arduino board through the I2C protocol. This protocol used SDA (data) and SCL (clock) lines which allowed for minimal wiring but still maintained efficient data transfer. Together, these communication methods turned out to be the backbones of the robot in a sense.

Software Design & Coding - Bhatnagar



The most important part of the system is the navigation logic, which uses the real time position data to travel towards the desired target. The code determines the direction using $\text{atan2}(y_direction, x_direction)$ and calculates the distance using $\sqrt{(x_direction * x_direction) + (y_direction * y_direction)}$. The function `calculateSmallestAngle()` makes the robot turn the shortest direction by adjusting the angle differently to remain within -180° to 180° .

The MQTT communication is used by continuously calling `mqtt_rebound()` and `client.loop()` to maintain the connection. The `callback()` function splits the data using `substring()` and converts values using `toFloat()` to update `x_robot`, `y_robot`, and orientation.

The obstacle avoidance system uses three ultrasonic sensors with `us_front.read(CM)`, `us_left.read(CM)`, and `us_right.read(CM)` to measure the distances. Conditional statements are used to allow the robot to react in real time by turning away from the obstacles and correcting its path. We used the logic of a previous assignment called the left hand loop to make the obstacle avoidance faster when the right or left sensors are less than 7.5 inches of distance, providing much more speed near the obstacles.

The motor control system uses `motor1.write()` and `motor2.write()`. Different values control the speed and direction of the motors, while the `stopMotors()` function sets both motors to 90 to stop the robot entirely.

The state machine logic organizes the progression between each target. When distance < 100 , the robot stops, waits 2 seconds using delay(2000) , and increments the target_index to move to the next target.

Code Screenshot:

Setup

```
void loop() {
  unsigned long stamp = millis();
  mqtt_rebound();
  client.loop();
  float x_direction = x_targets[target_index] - x_robot;
  float y_direction = y_targets[target_index] - y_robot;
  float targetAngle = atan2(y_direction, x_direction) * 180.0 / PI;
  float distance = sqrt((x_direction * x_direction) + (y_direction * y_direction));
  float diffAngle = calculateSmallestAngle(robotAngle, targetAngle);
  int t = 0;
  double df = us_front.read(CM) * 10 / 25.4;
  delay(5);
  double dl = us_left.read(CM) * 10 / 25.4;
  delay(5);
  double dr = us_right.read(CM) * 10 / 25.4;

  // Navigation State Machine
  if (target_index < 4) {
    // Stops when distance less than 100
    if (distance < 100) {
      stopMotors();
      delay(2000);
      target_index++;
    }

    if (diffAngle > 25 && dl > 4 && df > 5) {
      motor1.write(110);
      motor2.write(110);
    } else if (diffAngle < -25 && dr > 4 && df > 5) {
      motor1.write(70);
      motor2.write(70);
    } else {
      if (df < 4.5) {
        motor1.write(40);
        motor2.write(40);
        delay(270);
        if (df > 4 && dl > 2) {
          motor1.write(70);
          motor2.write(110);
          delay(360);
        }
      }
    }
  }
}
```

Loop (part 1)

```
// SIDE CORRECTIONS
if (dl < 7.5) {
  while (dl < 7.5 && df > 5) {
    df = us_front.read(CM) * 10 / 25.4;
    delay(5);
    dl = us_left.read(CM) * 10 / 25.4;
    delay(5);
    dr = us_right.read(CM) * 10 / 25.4;
    if (dl < 1.3) {
      motor1.write(110);
      motor2.write(70);
      delay(200);
    }
    if (dl < 7.5) {
      motor1.write(20);
      motor2.write(105);
    }
  }
}

if (dr < 7.5) {
  while (dr < 7.5 && df > 5) {
    df = us_front.read(CM) * 10 / 25.4;
    delay(5);
    dl = us_left.read(CM) * 10 / 25.4;
    delay(5);
    dr = us_right.read(CM) * 10 / 25.4;
    if (dr < 1.3) {
      motor2.write(110);
      motor1.write(70);
      delay(200);
    }
    if (dr < 7.5) {
      motor1.write(75);
      motor2.write(140);
    }
  }
}
```

```
while (distance > 250 && dr > 11 && dl > 8 && df > 13 && t < 10) {
  x_direction = x_targets[target_index] - x_robot;
  y_direction = y_targets[target_index] - y_robot;
  targetAngle = atan2(y_direction, x_direction) * 180.0 / PI;
  distance = sqrt((x_direction * x_direction) + (y_direction * y_direction));
  diffAngle = calculateSmallestAngle(robotAngle, targetAngle);

  df = us_front.read(CM) * 10 / 25.4;
  delay(5);
  dl = us_left.read(CM) * 10 / 25.4;
  delay(5);
  dr = us_right.read(CM) * 10 / 25.4;
  motor1.write(30);
  motor2.write(150);
  t++;
  if (t >= 10) {
    t = 0;
  }
}

if (distance > 100) {
  motor1.write(70);
  motor2.write(110);
}
} else {
  stopMotors();
}
```

Loop (part 2)

Assembly Instructions: EXPLODED VIEW W/ Bill of Materials - Bhatnagar

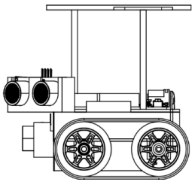
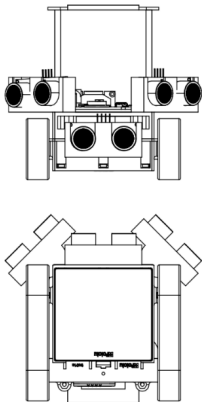
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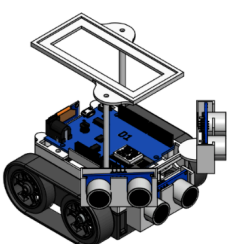
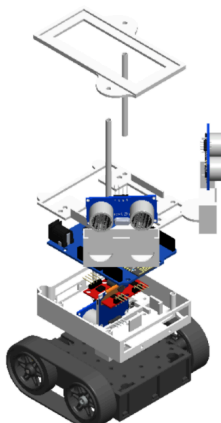
3

2

1

ITEM NO.	PART NUMBER	QTY.
1	Bottom Base	1
2	ZumoChassis	1
3	Sensor Holder	1
4	Top Base	1
5	WeMosD1Board	1
6	OLED Display	1
7	HCSR04	3
8	FeeTech-SMC-4CH	1
9	Marker Platform Rod	2
10	Marker Platform	1





NOTES:

NAME: Team J3

CLASS: SIT ENGR122 J1

PROJECT: Final Project Assembly Drawing

MATERIAL:

UNITS: INCH

MASS:

SCALE: 1:2

REV:

SIZE DWG. #

B

1

SHEET 1 OF 1

I pledge my honor that I have abided by the Stevens Honor System -- Aryan Bhatnagar

Last updated: Sunday, April 26, 2026 10:32:28 PM

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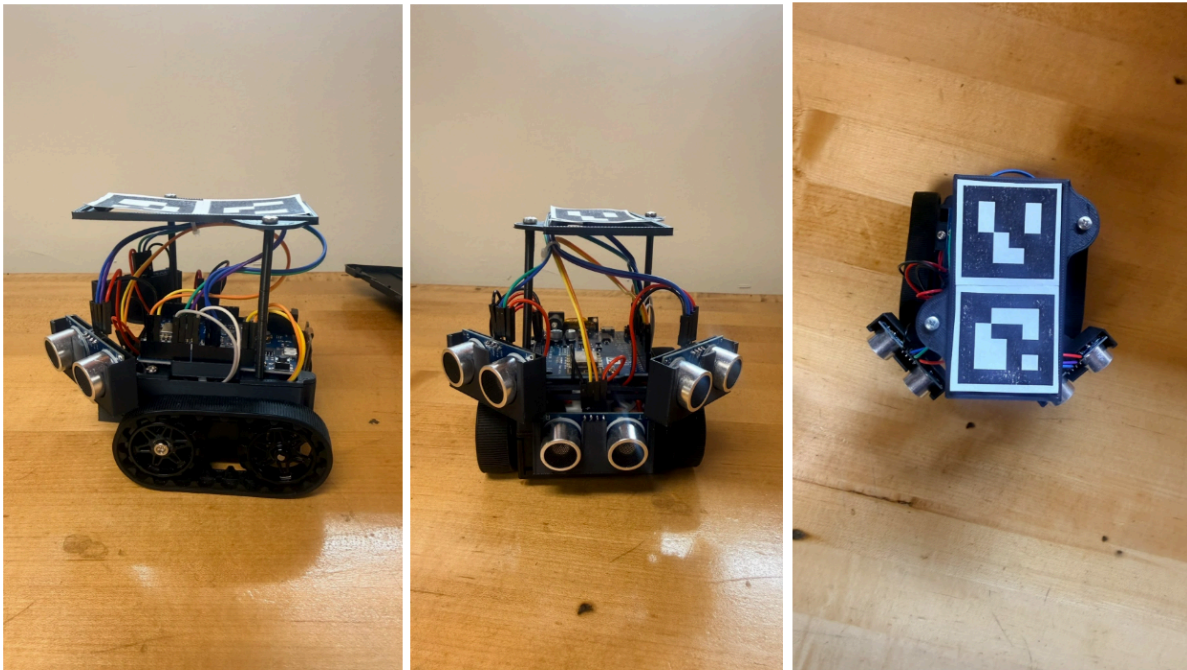
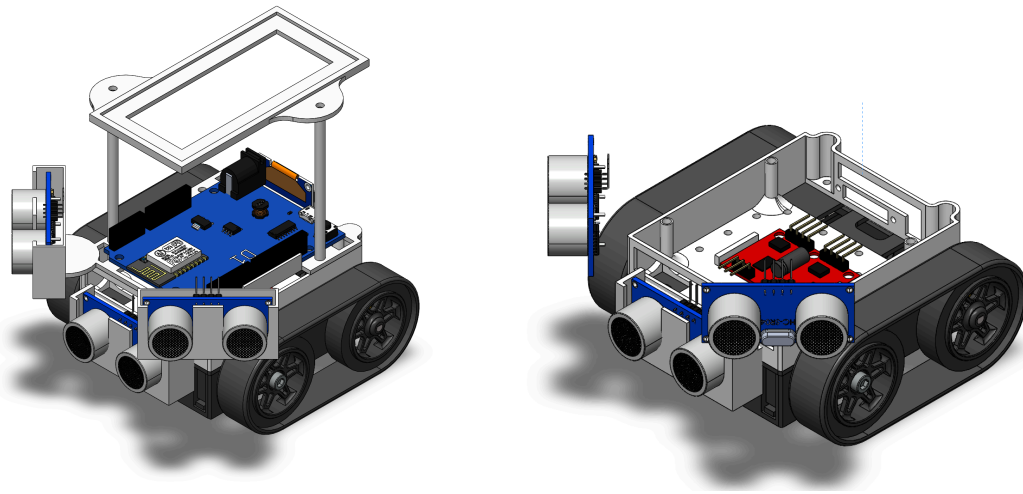
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3

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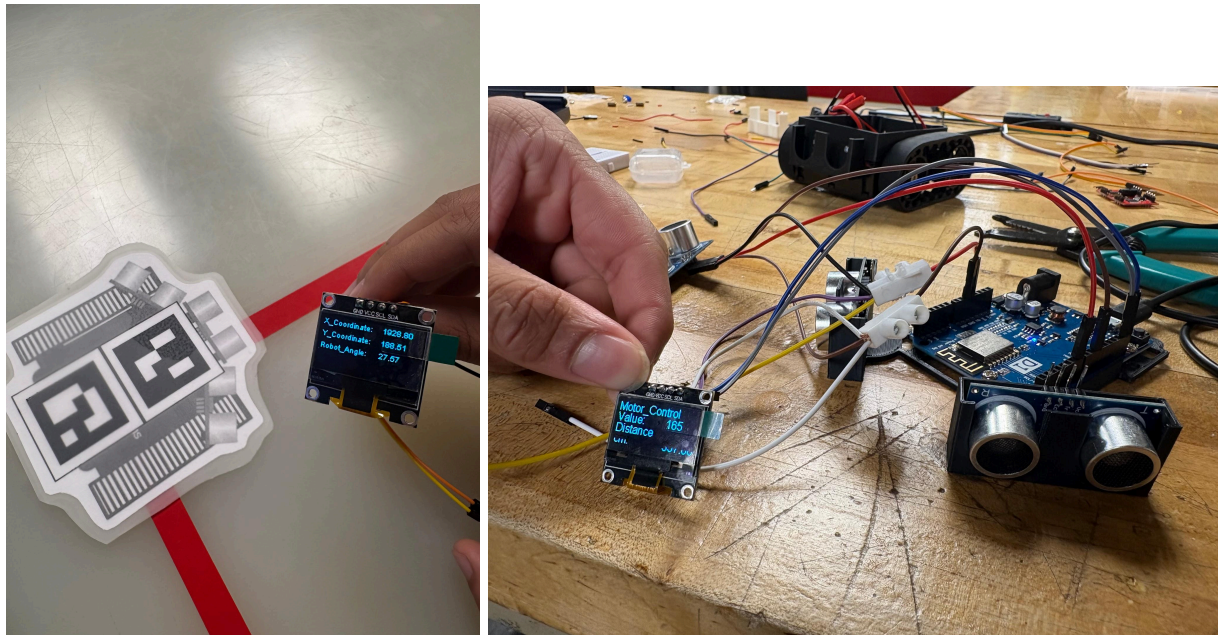
Prototype Specifications - Bhatnagar



Our prototype was built on a Zumo chassis, with a 3D- printed base supporting the Wemos D1R2 board, motor driver, and wiring. Ultrasonic sensors were mounted at the front for obstacle detection. The final design used the three ultrasonic sensors in front and on both sides of the robot to detect obstacles. This allowed the robot to react to nearby objects while following the target pathing. An OLED display was added to the back of the base. The OLED display was mounted onto the robot so we could debug and test while getting accurate information. It showed sensor and position values. Two vertical rods support the ArUco marker platform for position tracking. This was so the overhead camera system would be able to track the robot's position during navigation.

This whole solidworks assembly served as a way to make sure all the components of the robot are precision fit and there is as much margin or error as possible to make sure they actually fit. The assembly includes all the parts as shown except for the power bus. It was intentionally not created because we wanted to first wire everything and see where the power bus would end up. Because of this we decided to cut out a lot of holes on the bottom base so the power bus could be screwed in anywhere on the robot.

Performance Reporting and Project Management - Garcia



Throughout the project our team tested the robot through repeat trials. Early testing was focused on confirming the calibration of the ArUco markers (left image) by using the OLED display to cross reference the actual position shown on the camera to the position being read. This helped to verify that the MQTT position data would be accurate enough for navigation. We also tested the ultrasonic sensors and motor control systems to make sure that the robot could detect obstacles and respond properly. (right image). Based on these tests we adjusted the sensor thresholds and motor speed values to improve the robot's movement. Over time this helped to reduce unnecessary corrections that would result in us losing a large amount of time in the practice runs. With this we were able to improve obstacle avoidance and make the robot reach all four targets more efficiently.

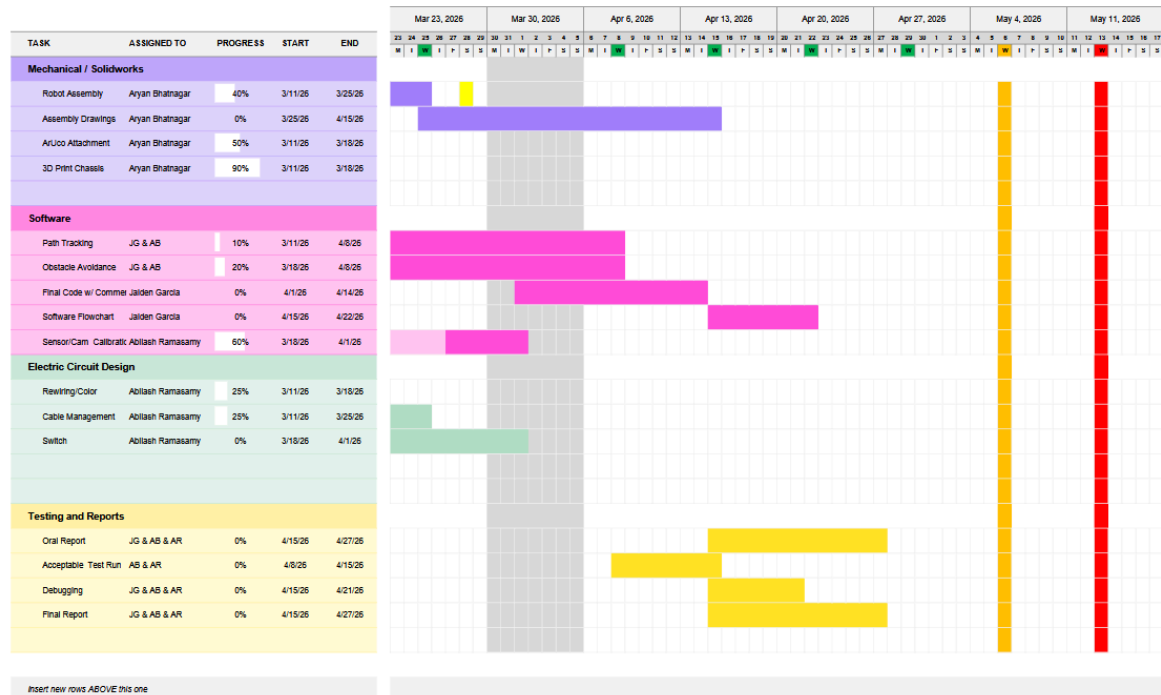
After constant iteration and tweaking of our software development, we ended up finishing our final run with a time of 51 seconds. The robot went through the course without any collisions and earned us 214.5 points.

ENGR 122 - J1

Jaiden Garcia, Aryan Bhatnagar, Abilash Ramasamy

Project start: Sun, 3/22/2026

Display week: 1



The work was divided equally between the three members. Each member was responsible for a certain design topic, but we all equally contributed in helping each other when needed. Originally in the Gantt chart, Aryan was in charge of the Mechanical Design. Abilash was in charge of the Electric Circuit design. Jaiden was in charge of the Software Design. However, by the end of the design process we helped each other equally. The group was able to communicate their problems well and get work done on time.

Conclusion and Recommendation - Garcia

Overall, the final robot design successfully met all project requirements as it demonstrated fast, reliable, and collision free autonomous navigation. However, several improvements could have been made to enhance performance. Our tread alignment was off and needed to be accounted for in our software development. Moreover, design could benefit mechanically from weight reduction and more simple printed parts. Also, we could make our ArUco stand more rigid as it was flimsy and broke during testing. Sensor placement could also be slightly adjusted to improve coverage. Our navigation system could be refined to create a smoother trajectory and get rid of unnecessary corrections. If the project requirements were to be expanded, then incorporating flexible speed control would be a possible enhancement.